

Biodiversity and Conservation

1. According to IUCN red list (2004), how many species of plants were documented as extinct species?

- (1) 359
- (2) 338
- (3) 87
- (4) 78

2. Read the following statements

Statement A : India is one of the 12 mega diversity countries of the world.

Statement B : Nearly 45,000 species of plants and triple as many of animals have been recorded from India.

- (1) Statement A is correct but Statement B is incorrect.
- (2) Statement A is incorrect but Statement B is correct.
- (3) Both statements are correct.
- (4) Both Statements are incorrect.

3. How many are example of exsitu conservation.

Zoological park, Botanical gardens, wildlife safari park, seed bank, National parks, Biosphere reserves, Sanctuaries

- (1) Seven
- (2) Three
- (3) Two
- (4) Four

4. Which one of the following cause of biodiversity loss, is because of a positive ecological interaction :

- (1) over-exploitation
- (2) Habitat fragmentation
- (3) Co-extinction
- (4) Alien species invasion

5. Match the following column correctly :

Column-A	Column-B
(A) Rare species	(i) Red panda
(B) Vulnerable species	(ii) Pigmy hog
(C) Endangered species	(iii) Arctic fox
(D) Critically endangered	(iv) Black buck
(1) A-i, B-ii, C-iii, D-iv	
(2) A-ii, B-iii, C-i, D-iv	
(3) A-iii, B-i, C-iv, D-ii	
(4) A-iii, B-iv, C-i, D-ii	

6. How many are the way of insitu conservation ?

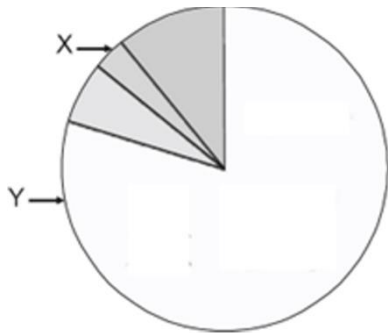
- (A) Germplasm collection
- (B) Sacred forests
- (C) Botanical gardens
- (D) Biosphere reserves
- (E) Safari Park
- (F) Sanctuaries

- (1) 2
- (2) 3
- (3) 4
- (4) 5

7. Read the following statements and select the incorrect one.

- (1) Total number of biodiversity hotspot in the world is 34.
- (2) At the ecosystem level, India has a greater ecosystem diversity than a Scandinavian country like Norway.
- (3) Amazon rain forest is estimated to produce 20 percent of the total oxygen in the earth's atmosphere.
- (4) Biodiversity hotspots are characterised by high levels of species richness and lack of endemism.

8. Observe the pie chart given below showing the global diversity regarding total number of the invertebrate species. Select the correct option for X and Y



- (1) X-Molluscs, Y- Fishes
(2) X-Crustaceans, Y- Insects
(3) X-Insects, Y- Molluscs
(4) X-Crustaceans, Y- Molluscs
9. Tropical regions show greatest level of species richness because
- A. Tropical latitudes have remained relatively undisturbed for millions of years, hence more time was available for species diversification.
B. Tropical environments are more seasonal.
C. More solar energy is available in tropics.
D. Constant environments promote niche specialization.
E. Tropical environments are constant and predictable.

Choose the correct answer from the options given below.

- (1) A, C, D and E only
(2) A and B only
(3) A, B and E only
(4) A, B and D only
10. Which one of the following has the greatest biodiversity on earth?
- (1) Tropical Amazonian rain forest
(2) Eastern ghats
(3) Western ghats
(4) All of the above

11. Identify the incorrect statement w.r.t. biodiversity hotspots

- (1) When all the biodiversity hotspots put together, they cover less than 2% of earth's land area
(2) Initially 25 biodiversity hotspots were identified
(3) They are the richest and least threatened reservoir of plant and animal life on earth
(4) Strict protection of the hotspots can reduce the ongoing mass extinction by almost 30%

12. In species - Area relationship the value of z depends on which of the following factors ?

- (a) Species richness
(b) Area
(c) Taxonomical category the species belongs

- (1) (a) alone
(2) (b) alone
(3) (a) and (b)
(4) (a), (b) and (c)

13. Which of the following is not correct ?

- (1) Gir National Park - Asiatic lion - Gujarat
(2) Corbett National Park - Elephant - Maharashtra
(3) Kaziranga National Park - Rhinoceros - Assam
(4) Kanha National Park - Tiger - Madhya Pradesh

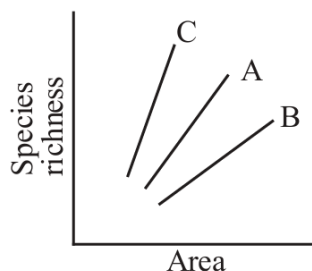
14. Which one of the following group of organisms are exotic species introduced in India ?

- (1) Lantana, Ekhhornia, Vallisneria
(2) Nelumbium, Parthenium, Ekhhornia
(3) Lantana, Eichhornia, Parthcnium
(4) Nelumbium, Nymphaca, Eichhornia

15. Tropical areas are having greatest biodiversity because :

- (1) They had a long undisturbed evolutionary time
(2) They are less seasonal
(3) More solar energy is available in tropics
(4) All of these

16. According Alexander Von Humbolt, in the graph of Species Area relation of Area A, B and C which of the following area has steeper slope and maximum species richness respectively



- (1) A & B
- (2) C & A
- (3) C & B
- (4) C & C

17. Which of the following regions of our country are known for their rich biodiversity?

- (1) Western ghats and eastern himalayas
- (2) Western ghats and deccan plateau
- (3) Eastern himalayas and gangetic plane
- (4) Trans Himalayas and deccan peninsula

18. Read the following statements and select the correct option.

Statement A: A stable community should not show too much variation in productivity from year to year.

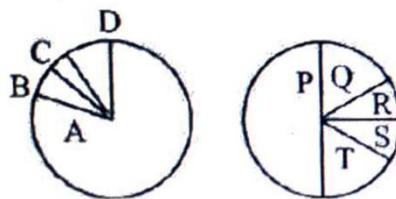
Statement B: The relation between species richness and area for a wide variety of taxa turns out to be rectangular hyperbola.

- (1) Only statement A is correct
- (2) Only statement B is correct
- (3) Both statements A and B are correct
- (4) Both statements A and B are incorrect

19. Dachigam National Park, Jammu and Kashmir is home to the famous

- (1) Musk deer
- (2) Tiger
- (3) Lion
- (4) Rhino

20. Below given pie diagram represent global Biodiversity of invertebrates and vertebrates:



"A" and "P" respectively are

- (1) Molluscs and Birds
- (2) Molluscs and Reptiles
- (3) Insects and Reptiles
- (4) Insects and Fishes

21. Read the following statements and select the correct option.

Statement A: Robert May places the Indian species diversity at about 7 million.

Statement B: Among animals, insects are the most species-rich taxonomic group, making up more than 70 per cent of the total.

- (1) Statement A is correct but Statement B is incorrect.
- (2) Statement A is incorrect but Statement B is correct.
- (3) Both statements are correct.
- (4) Both Statements are incorrect.

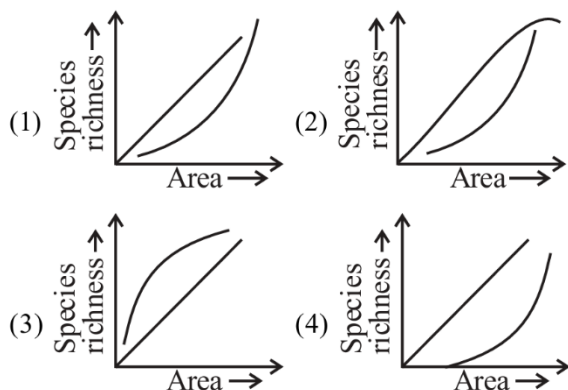
22. So far 1.5 million species are identified, in which the number of fungi species identified is more than the combined total of:

- (1) Algae, lichens, mosses and ferns
- (2) Fishes, amphibians, reptiles and mammals
- (3) Molluscs and crustaceans
- (4) Molluscs, fishes and amphibians

23. For frugivorous birds and mammals in the tropical forests of different continents. Z (slope of the line/regression coefficient) is found to be

- (1) 1.15
- (2) 1
- (3) 0.2
- (4) 0.1

24. Identify species area relationship graph :



25. More conservative and scientifically sound estimate made by Robert May place global species diversity at about.....:

- (1) 70 million
- (2) 7 billion
- (3) 7 lac
- (4) 7 million

26. Read the following statements Assertion (A):

Biologists are not sure about how many prokaryotic species there might be.

Reason (R): Conventional taxonomic methods are not suitable for identifying microbial species

- (1) Both (A) and (R) are true and (R) is correct explanation of (A)
- (2) Both (A) and (R) are true but (R) is not the correct explanation of (A)
- (3) (A) is true but (R) is false
- (4) Both (A) and (R) are false

27. Match the columns I and II with respect to the country and the number of birds found there.

Column I	Column II
A. Colombia	P. 1,400 species of birds
B. India	Q. 105 species
C. New York	R. 1,200 species
D. Greenland	S. 56 species

- (1) A- R , B- P , C- Q , D- S
- (2) A- S , B- R , C- P , D- Q
- (3) A- P , B- R , C- Q , D- S
- (4) A- Q , B- P , C- R , D- S

28. Many species extinctions in last 500 years were due to overexploitation by humans. This is true for which of the following species?

- (1) Nile perch
- (2) Eichhornia
- (3) African catfish
- (4) Steller's sea cow

29. Given below are two statements

Assertion: If the present trends of species losses continue, then nearly half of all the species might be wiped out within the next 200 years.

Reason: The accelerated rates of species extinctions that the world is facing now are largely due to volcanic activities.

- (1) Both Assertion & Reason are true and the reason is the correct explanation of the assertion
- (2) Both Assertion & Reason are true but the reason is not the correct explanation of the assertion
- (3) Assertion is true statement but Reason is false
- (4) Both Assertion and Reason are false statements

30. Introduction of *Clarius gariepinus* in aquaculture in India is a serious threat to extinction of

- (1) Cray fish
- (2) Cichlidfish
- (3) Native cat fish
- (4) Salmon fish

31. Which of the following statements is correct?

- (1) Species diversity in general increases from poles to equator
- (2) Conventional taxonomic methods are equally suitable for higher plants and microorganisms
- (3) India's share of global species diversity is about 18 %
- (4) There are about 25000 known species of plants in India

32. Read the following statements:

I. Species diversity decreases as we move away from the equator towards the poles.

II. A forest in a temperate region has up to 10 times as many species of vascular plants as a forest of equal area in a tropical region.

III. Tropical environments, unlike temperate ones, are less seasonal, relatively more constant and predictable.

Regarding pattern of biodiversity, the correct are

- (1) I and II only
- (2) II and III only
- (3) I and III only
- (4) All I, II and III

33. Choose the correct option of percentage of ascending order of given living beings species which faces threat of extinction at presently-

- (1) Amphibians, Gymnosperms, Mammals, Birds
- (2) Birds, Mammals, Gymnosperms, Amphibians
- (3) Birds, Amphibians, Mammals, Gymnosperms
- (4) Mammals, Birds, Gymnosperms, Amphibians

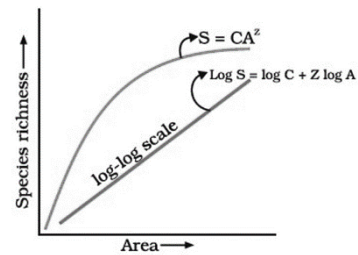
34. Global species diversity and land area covered by India with respect to world is respectively:

- (1) 12%, 7%
- (2) 2.4%, 9%
- (3) 8.1%, 2.4%
- (4) 4%, 3%

35. Extinction is a naturally occurring phenomenon that has occurred on earth, since life began, scientists are so concerned about extinction now because

- (1) Human activities are directly responsible for today's mass extinctions
- (2) Extinction is occurring faster than ever before and faster than rates of restoration
- (3) Large number of plant species which form the base of terrestrial food webs become extinct
- (4) All of these

36. Which of the following is correct ?



- (1) $S =$ exponential growth, $C =$ Y intercept. Z Unknown factor, $A =$ area
- (2) $S =$ species richness, $C =$ Y intercept, Z regression coefficient, $A =$ area
- (3) $S =$ Logistic growth, $C =$ carrying capacity $Z =$ Environmental resistance, $A =$ area
- (4) $S =$ Semi arid species, $C =$ Y intercept. $Z =$ regression coefficient, $A =$ area

37. The term that describes the species that are at risk of extinction in the wild in the near future is

- (1) Extinct species
- (2) Endangered species
- (3) Vulnerable species
- (4) Endemic species

38. The historic convention "The Earth Summit" held in 1992 was called upon all the nations to take appropriate measures for

- (1) Conservation of biodiversity and sustainable utilisation of its benefits
- (2) Reducing greenhouse gas emission and global warming
- (3) Reducing release of persistent organic pollutant in atmosphere
- (4) Ex situ conservation of endangered species of wild flora and fauna

39. In the given equation of species-area relationship $\log S = \log C + Z \log A$, 'Z' stands for

- (1) Species richness
- (2) Regression coefficient
- (3) Y-intercept
- (4) Area of the habitat

40. Regarding the number of birds species inhabiting an area, which of the following representation is correct?

- (1) New York > India > Greenland
- (2) Colombia > Greenland > India
- (3) India > New York > Colombia
- (4) Colombia > New York > Greenland

41. Read the following statements and state them as true (T) or false (F).

A. India with only 2.4% of the world's land area possesses 8.1% species diversity of the world.

B. Only about 50,000 genetically different strains of rice are found in whole world.

C. The biodiversity decreases when we move from low to high latitude.

- (1) A – T , B – T , C – T
- (2) A – T , B – F , C – T
- (3) A – T , B – F , C – F
- (4) A – F , B – F , C – F

42. Regarding the causes of biodiversity losses, which one of the following is not considered amongst 'The Evil Quartet'?

- (1) Alien species invasions
- (2) Over-exploitation
- (3) Co-evolution
- (4) Co-extinctions

43. The reasons behind conserving biodiversity can be grouped into categories, which include

- I. Broadly utilitarian**
- II. Narrowly utilitarian**
- III. Long utilitarian**
- IV. Ethical utilitarian**
- V. Small utilitarian**

Choose the correct option.

- (1) I, II, III and IV
- (2) II, III and IV
- (3) I, II and IV
- (4) I, III, IV and V

44. Consider following strategies for conservation of biodiversity and select correct option for in-situ approaches

- i. Sacred groves**
- ii. Wild life safari**
- iii. National parks**
- iv. Tissue culture**
- v. Germplasm storage**
- vi. Biosphere reserve**

- (1) ii, iii and vi
- (2) i, iii and vi
- (3) i, iv, v and vi
- (4) i, ii, iii and vi

45. Find the wrongly matched pair.

- (1) Endemism - Species confined to one region and also found in other regions
- (2) Alien species - *Clarias gariepinus*
- (3) Lungs of the planet - Amazon rain forest
- (4) Hot spots - Regions with species richness

46. What is common to the techniques

- (i) in vitro fertilisation**
- (ii) Cryopreservation**
- (iii) tissue culture**

- (1) All are in situ conservation methods.
- (2) All are ex situ conservation methods.
- (3) All require ultra modern equipment
- (4) All are methods of conservation of extinct organisms.

47. Read the following statements

Assertion- Red Data Book categorises all the plant species occurring on the earth.

Reason - Blue whale, black buck and snow leopard are some animals mentioned in Red Data Book.

- (1) Both Assertion & Reason are true and the reason is the correct explanation of the assertion
- (2) Both Assertion & Reason are true but the reason is not the correct explanation of the assertion
- (3) Assertion is true statement but Reason is false
- (4) Both Assertion and Reason are false statements

48. State the following statements as true (T) or false (F) and select the correct option.

A. Genetic diversity enables a population to adapt to its environment.

B. Among the animals, insects are the most species rich taxonomic group.

C. Tropical amazonian rain forest has the greatest biodiversity on earth.

(1) A – F , B – T , C – F

(2) A – T , B – F , C – T

(3) A – T , B – T , C – T

(4) A – F , B – F , C – F

49. Which of the following species is/are extinct in the last 500 years due to over exploitation of natural resources :

(1) Stellar's sea cow

(2) Passenger pigeon

(3) Water hyacinth

(4) Both (1) and (2)

50. Which of the following statements are correct regarding fragmentation of habitats?

I. It results in disruption of complex interactions amongst species

II. It affects the animals requiring large territories

III. It causes destruction of species restricted to deeper undisturbed parts of forests

IV. Fragmented areas have reduced biodiversity

(1) I and II only

(2) II and III only

(3) I, III and IV only

(4) I, II, III and IV

Answer Key

1. (3)
2. (1)
3. (4)
4. (3)
5. (4)
6. (2)
7. (4)
8. (2)
9. (1)
10. (1)
11. (3)
12. (3)
13. (2)
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16. (4)
17. (1)
18. (3)

19. (1)
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45. (1)
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47. (4)
48. (3)
49. (4)
50. (4)



